



MD Mohimul Alam

Computer Science Engineer — Full-Stack Developer — Blockchain
Researcher — Cybersecurity Enthusiast — Springer-Published Au-
thor

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🌐 Portfolio

🐙 GitHub

🎓 Google Scholar

PROFESSIONAL PROFILE

- Frontend developer with 2+ years of experience in React.js, Next.js, TypeScript, and modern UI/UX design.
- Expert in building responsive dashboards, e-commerce platforms, and interactive web applications with real-time updates.
- Blockchain researcher specializing in Hyperledger Fabric, Zero-Knowledge Proofs, and secure distributed architectures.
- Full-stack development experience with Node.js, REST APIs, and production-ready applications.
- Passionate about creating high-performance, user-friendly, and visually appealing web solutions.
- Portfolio: mohim-portfolio.netlify.app

EDUCATION

Independent University, Bangladesh

2019–2025

B.Sc. in Computer Science & Engineering

CGPA: 2.73

TECHNICAL SKILLS

- **Frontend:** React.js, Next.js, TypeScript, TailwindCSS, JavaScript, HTML5, CSS3, Figma, Framer Motion, Responsive Design, Redux, Zustand
- **Backend:** Node.js, Express.js, REST API, GraphQL, MongoDB, MySQL, JWT Authentication, Firebase
- **Blockchain & Security:** Hyperledger Fabric, Smart Contracts, Zero-Knowledge Proofs, Threshold Encryption, Formal Verification
- **Testing & QA:** Jest, React Testing Library, Unit Testing, Test-Driven Development, SonarQube
- **Tools & Platforms:** Git, GitHub Actions, Docker, AWS (S3, CloudFront, ALB), Vite, Postman, Linux, CI/CD
- **Specializations:** IoT Development (ESP32), Embedded Systems, Real-time Applications, Performance

PORTFOLIO & KEY PROJECTS

- **Advanced Blockchain Voting Framework** *Hyperledger Fabric, Zero-Knowledge Proofs, Threshold Encryption, React, Node.js*
 - Designed and implemented an enhanced blockchain-based electronic voting framework with advanced cryptographic protocols.
 - Integrated Zero-Knowledge Proofs and Threshold Encryption to strengthen voter privacy, verifiability, and distributed trust.
 - Improved transaction throughput by approximately 45% and reduced operational latency by 38% through protocol and architecture optimization.
 - Developed a production-ready platform supporting secure, scalable, and privacy-preserving electronic elections.
 - Research published in *Springer SN Computer Science* (2026): “An Extended Formal Framework for Blockchain-Based Voting Systems with Advanced Cryptographic and Performance Enhancements.”
[Springer Publication](#)
- **SonnyNY E-Commerce Platform** *React.js, TailwindCSS, CJ Dropshipping API, AWS*
 - Built a full-featured e-commerce platform with real-time product data from the CJ Dropshipping API
 - Implemented dynamic category navigation, product listings, cart, wishlist, and checkout functionality
 - Integrated authentication guards with Redux a for secure shopping experience
 - Optimized performance with caching, lazy loading, and parallel API requests
 - Deployed to AWS with ALB, S3, and CloudFront; set up CI/CD pipeline with GitHub Actions [GitHub Repository](#)
- **Blockchain-Based Voting System** *Hyperledger Fabric, React, Node.js*
 - Enterprise-grade voting platform with 4 organizations, ,secure encrypted voting, and real-time auditing
 - Implemented Zero-Knowledge Proofs and Threshold Encryption for voter privacy
 - Published paper: “Blockchain-Based Voting System with Hyperledger Fabric and React Interface” — Springer CCIS, ICDSAIA 2025 [GitHub Repository](#)
- **ESP32-Based Health Monitoring System** *ESP32, MAX30102, AD8232, Hyperledger Fabric*
 - Real-time patient vitals monitoring using ECG and SpO₂ sensors integrated with an ESP32 micro-controller
 - Secure data pipeline from edge device to Hyperledger Fabric blockchain with AES-128 encryption and RBAC
 - Achieved 95.4% ECG and 93.8% SpO₂ accuracy with 98% system uptime in clinic field trials
 - Optimized hardware design delivering 72+ hours of battery-backed operation at BDT 2,800–4,000 per unit [GitHub Repository](#)
- **Healthcare Data Exchange Network** *Hyperledger Fabric, React, Node.js*
 - Private blockchain for 3 healthcare institutions with granular RBAC and real-time dashboard
 - Ensured 98% data consistency in clinical simulations [GitHub Repository](#)
- **Odd Studio Vision Website** *Next.js, TailwindCSS, React*
 - Built a creative agency website for digital branding, videography, photography, and marketing strategy
 - Designed as both a portfolio and lead-generation platform with modern UI/UX
 - Optimized for performance, cross-device compatibility, and smooth navigation
 - Features interactive components, animations, and responsive design [Live Website](#)
- **E-Commerce Web Application** *React, JavaScript, Node.js, MySQL, TailwindCSS*
 - Developed responsive product catalog, shopping cart, and admin dashboard

- Integrated order management and simulated payment gateway with real-time updates [GitHub Repository 1](#) [GitHub Repository 2](#)
- **Santaran & MBTS Projects** *Nuxt.js, React, Vue.js*
 - Built responsive web platforms for artist-run organizations and event management
 - Implemented interactive dashboards, project management tools, and data visualization
 - Designed mobile-first responsive UI with animations and dynamic content [Santaran MBTS](#)

RESEARCH & PUBLICATIONS

- **An Extended Formal Framework for Blockchain-Based Voting Systems with Advanced Cryptographic and Performance Enhancements** — *SN Computer Science (Springer)*, Accepted May 2026
DOI: 10.1007/s42979-026-05127-w
- **Blockchain-Based Voting System with Hyperledger Fabric and React Interface** — Springer CCIS, ICDSAIA 2025
DOI: 10.1007/978-3-032-11352-8_22
- **Real-Time Patient Health Monitoring Using IoT Sensors and Blockchain Technology** — Research paper on ESP32-based health monitoring with ECG and SpO₂ sensors integrated into a Hyperledger Fabric blockchain network
- **Healthcare Data Exchange Network** — Paper in progress, focusing on secure private blockchain for hospitals

PROFESSIONAL EXPERIENCE

Shop Online New York Jan 2025 – Present

React Developer - Remote

- Developed and maintained a large-scale e-commerce platform using React.js, TailwindCSS, and Redux
- Integrated CJ Dropshipping API for real-time product data, categories, and inventory management
- Implemented dynamic navigation, product filtering, cart, wishlist, and checkout flows
- Optimized performance with lazy loading, caching, and parallel API requests
- Set up CI/CD pipeline with GitHub Actions for automated deployment to AWS (S3, CloudFront)
- Collaborated with cross-functional teams including product managers, designers, and backend engineers
- Wrote clean, maintainable code with TypeScript and Jest for unit testing

BetaFore - Software & Digital Media May 2024 – Oct 2024

React Developer Intern

- Built responsive e-commerce UI with product catalog, filtering, and search functionality
- Collaborated with the backend team for REST API integration and data synchronization
- Implemented modern UI/UX designs using TailwindCSS and responsive web design principles

TechTrioZ Solutions Jan 2023 – Feb 2024

Data Entry Specialist

- Managed digital data entry systems and geospatial data processing
- Maintained 99.8% data accuracy across large datasets

CERTIFICATIONS

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- Growth Hacking & Online Campaigns Internship — RIT & Excelerate (2025)
 - React.js & Node.js — Global Skills Development Agency (2023)
 - CCNAv7: Switching, Routing, and Wireless Essentials — Cisco (2024)
 - CCNAv7: Introduction to Networks — Cisco (2022)
 - Ethical Hacker — Cisco Networking Academy (2024)

LEADERSHIP & ACTIVITIES

- JCI Bangladesh — General Member (2023–Present)
 - Volunteered at community events and organized sports tournaments
 - Participated in leadership development programs and community service initiatives
- Bangladesh Red Crescent Society — Member (2014–Present)
 - Participated in disaster preparedness camps and humanitarian relief programs